

Multimodal RAG Benchmark Document

Synthetic 20-page technical PDF for testing retrieval across text, tables, diagrams and cross-modal evidence.

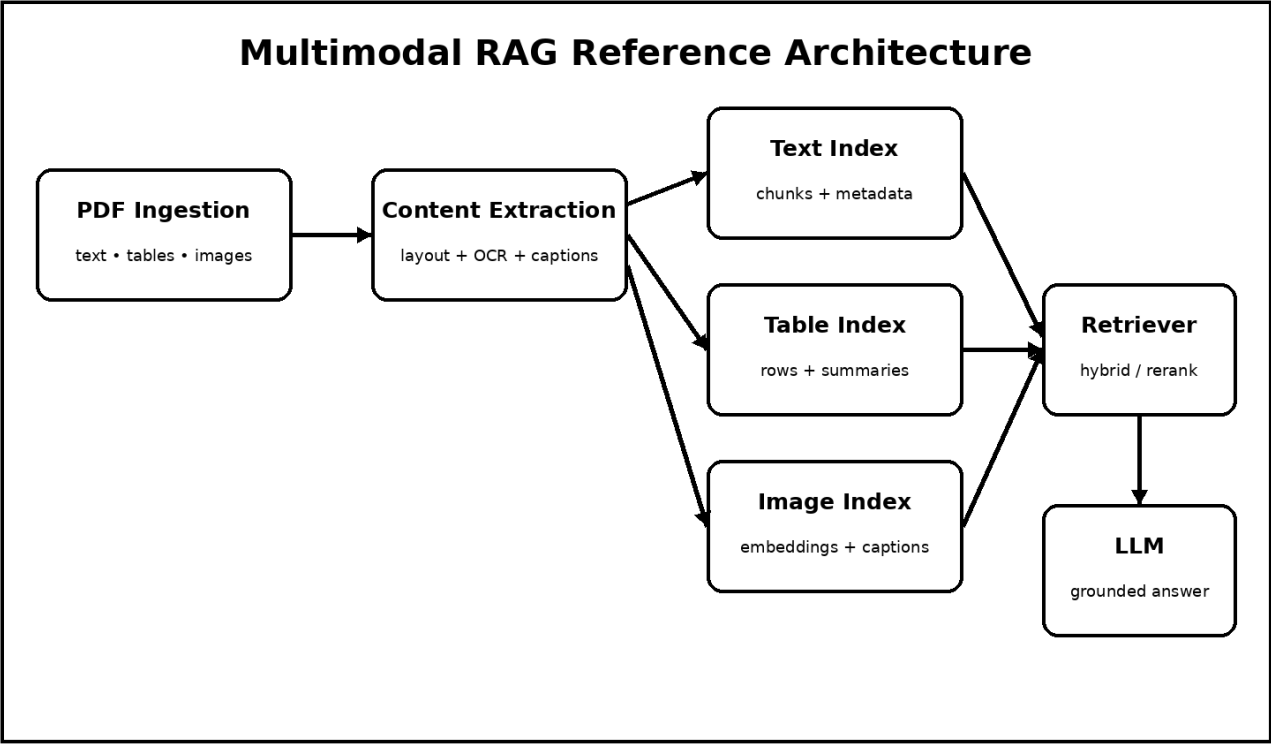


Figure 1. Reference architecture for the Aurora Claims Intelligence Platform.

Document ID: MRAG-TEST-001 **Version:** 1.0 **Domain:** Document Intelligence

This document is synthetic and intentionally contains information distributed across narrative sections, structured tables and visual diagrams. It is designed to expose weak chunking, missing metadata, table-context loss and diagram-retrieval failures.

1. Executive Overview

The Aurora Claims Intelligence Platform (ACIP) is a fictional enterprise system for insurance claim intake. It accepts PDF claim packets containing narrative descriptions, policy schedules, damage photographs, workflow diagrams and structured forms. Its retrieval layer preserves relationships between pages, text blocks, tables, images and generated captions.

The system recognizes five evidence classes: narrative text, tables, images, diagrams and metadata. Every extracted element receives a document identifier, page number, section identifier, modality label and confidence score. Related evidence is joined with an evidence_group_id.

Objective	Description	Target
Recall	Retrieve evidence needed to answer a question	≥ 0.90
Grounding	Factual claims supported by evidence	≥ 0.95
Modality coverage	Correctly retrieve text/tables/figures	≥ 0.90
Latency	Query to first grounded result	< 2.5 s
Traceability	Return page and evidence type	100%

2. Reference Architecture

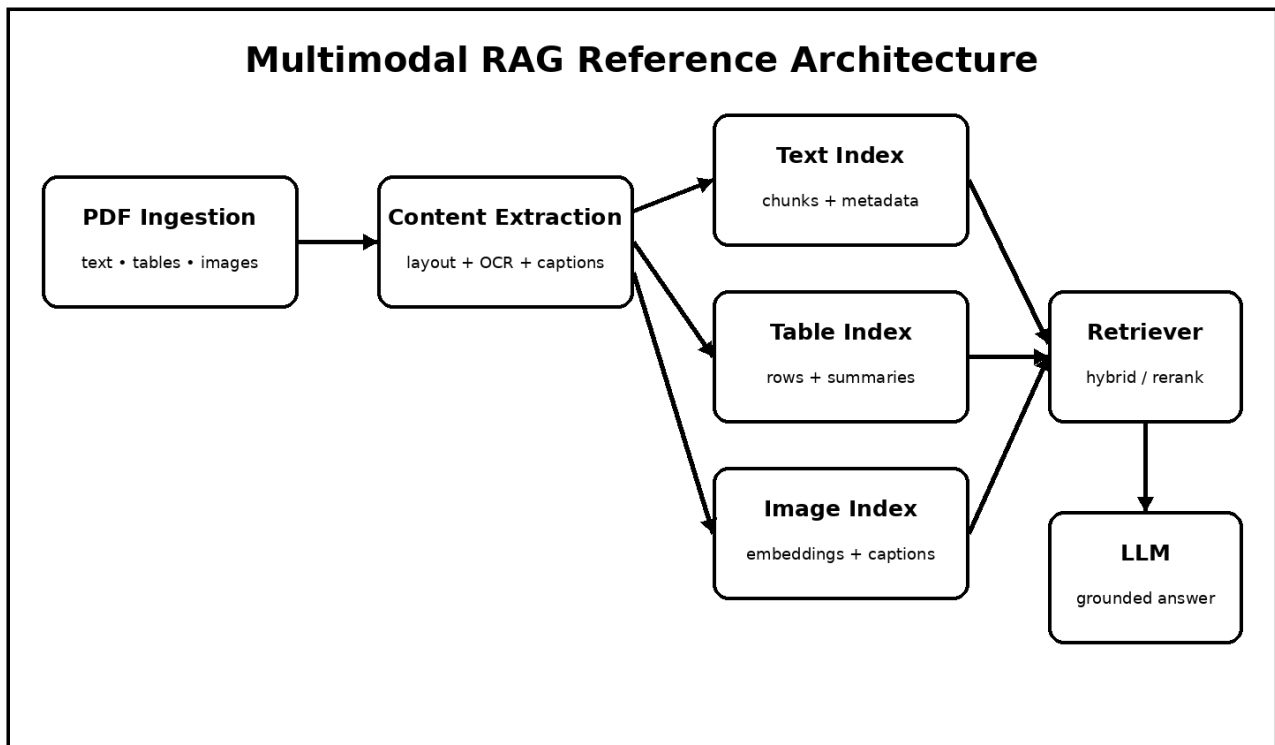


Figure 2. The reference architecture separates text, table and image evidence before retrieval.

The ingestion layer identifies layout blocks. Text is semantically chunked; tables are represented as structured rows plus summaries; images are stored with captions, OCR and visual embeddings. A hybrid retriever combines lexical and vector candidates before reranking.

Component	Responsibility	Metadata
PDF Ingestion	Preserve page/layout	doc_id, page_no, block_id
Content Extraction	Create modality evidence	modality, bbox, confidence
Text Index	Semantic + lexical retrieval	chunk_id, section_id
Table Index	Rows + summaries	table_id, row_id, header
Image Index	Visual retrieval	image_id, caption, bbox
Retriever	Merge/rerank evidence	scores, evidence_group_id

3. Document Ingestion and Normalization

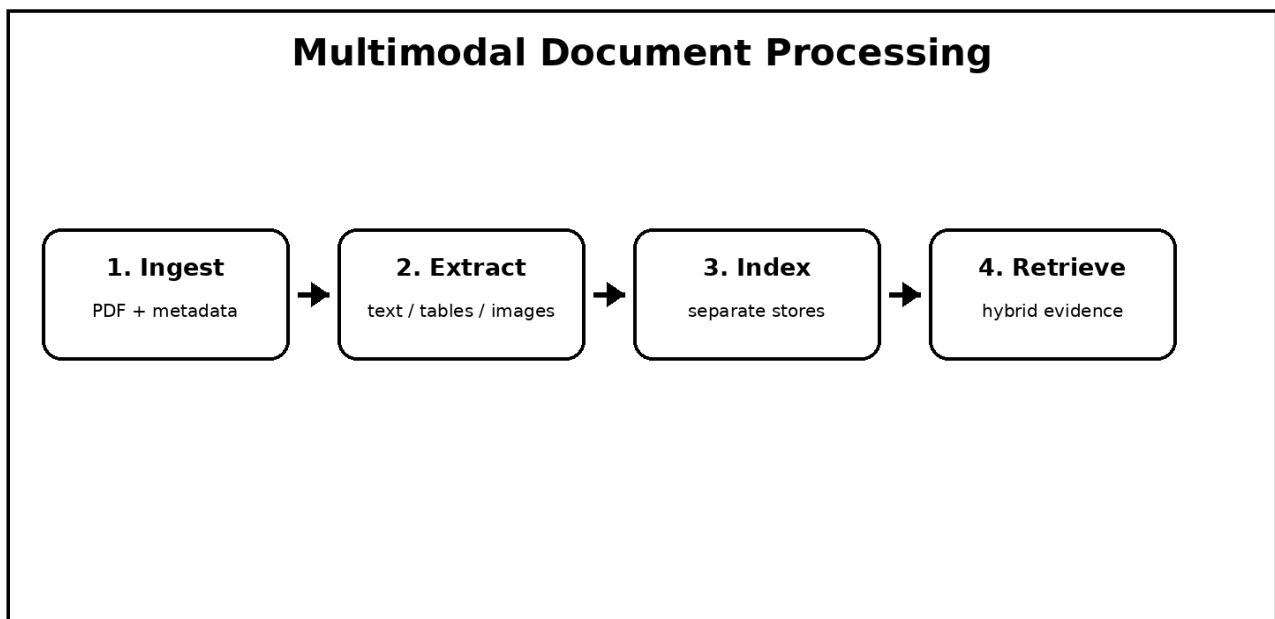


Figure 3. Ingestion order: ingest, extract, index, retrieve.

A PDF is registered with a stable document ID. The parser detects layout blocks. Text blocks retain headings because headings become hierarchical metadata. Tables preserve column headers. Images are stored separately and linked to nearby captions when layout indicates an association.

Rule	Condition	Action
Q1	Table header confidence < 0.85	Run table fallback parser
Q2	Image OCR confidence < 0.75	Keep image; mark OCR low confidence
Q3	Reading-order ambiguity	Use coordinates + heading hierarchy
Q4	Caption separated from image	Link by figure label and proximity
Q5	Duplicate page object	Deduplicate by page hash + overlap

4. Chunking and Metadata Strategy

ACIP uses semantic chunking rather than a fixed character window. A section heading starts a context boundary. Tables are never split without repeating headers. A diagram receives a figure-level record and optional region-level records.

Field	Example	Purpose
doc_id	MRAG-TEST-001	Document grouping
page_no	7	Source traceability
section_id	SEC-4.2	Hierarchical filtering
modality	table	Modality-aware retrieval
evidence_group_id	EG-042	Join related evidence
parent_id	TABLE-07	Structural relationship
bbox	[82,144,510,390]	Visual localization
confidence	0.96	Extraction quality

For tables, the preferred retrieval unit is a row enriched with its headers. A row such as *Commercial Property | Critical | 24h* must remain associated with headers such as *Claim Type, Priority, Target Response*. Returning the row alone creates ambiguity.

5. Operational Policy and Service Levels

ACIP classifies cases into Standard, Priority, Critical and Escalated lanes. The following table is authoritative for response targets. Narrative examples elsewhere are illustrative unless they explicitly refer back to this table.

Lane	Response target	Analyst review	Notification	Escalation
Standard	8 business hours	Within 4h	After validation	If target missed
Priority	4 business hours	Within 2h	Immediately after validation	If unresolved in 4h
Critical	60 minutes	Within 30m	Immediately	Mandatory supervisor review
Escalated	30 minutes	Within 15m	Immediately + supervisor	Engineering + operations

Table 4. Operational service levels.

Response target and analyst-review target are different concepts. A Critical case can require analyst inspection within 30 minutes while its response target is 60 minutes. Escalated is a workflow state rather than a severity level and may be reached from any lane.

6. Claim Evidence Model

A claim packet may contain narrative facts, policy context, visual evidence, workflow evidence and audit records. These objects are linked by `claim_id`, but retrieval should also support policy numbers, incident descriptions and visual concepts.

Evidence	Required fields	Typical source	Retrieval hint
Narrative	event_date, description, location	Email/body	Semantic + keyword
Policy	policy_no, start, end, deductible	Policy table	Exact + table
Visual	image_id, caption, observations	Photograph/figure	Visual + caption
Workflow	step, predecessor, successor	Flow diagram	Diagram retrieval
Audit	actor, action, timestamp	Audit table	Exact metadata

Example claim records

Claim ID	Policy	Incident	Severity	Lane
CLM-2047	POL-88142	Water ingress	High	Priority
CLM-2051	POL-77403	Vehicle impact	Critical	Critical
CLM-2059	POL-99112	Glass breakage	Medium	Standard
CLM-2064	POL-55021	Electrical fire	High	Escalated